

Igor Ignashin

Research Engineer / LLM Post-Training - distillation, SFT, evaluation, multi-agent RL

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Profile

M.S. candidate at MIPT working across LLM post-training, applied ML, and optimization research. Main applied focus: distillation/SFT, teacher-generated data, reward/grader parsing, benchmark evaluation, and multi-agent RL/NLHF.

At MIPT AI Institute / BRAIn Lab I worked as a Research Team Lead. In these projects my role was mixed: I set research direction and also worked hands-on on problem framing, code, experiments, and analysis. I combined leadership and hands-on work in distillation/SFT, multi-agent RL/NLHF, SGD dynamics, and optimization tracks; in LLM compression my role was a major hands-on contribution to framing and analyzing experiments.

- LLM post-training: SFT, teacher/KL distillation, synthetic and teacher-generated data.
- Evaluation: graders, reward parsing, benchmark tables, baseline and teacher comparison.
- Research leadership: distillation/SFT, multi-agent RL/NLHF, optimization tracks; major hands-on contribution in LLM compression.
- ML engineering: Python, C++, SQL, PyTorch, Hugging Face workflows, Linux, Git.

Experience

Research Team Lead / LLM Post-Training Engineer | MIPT AI Institute / BRAIn Lab / MIRAI; LAB MMO collaboration | 2024-2026

Led five small research teams of 1-3 people. In most projects the role was combined: led the research direction and also contributed technically by selecting baselines, decomposing experiments, writing/reviewing code, checking metrics, and turning results into reproducible artifacts.

- Qwen / DeepScaleR math distillation: built an end-to-end research pipeline for transferring part of the DeepScaleR-1.5B-Preview RL gain back into DeepSeek-R1-Distill-Qwen-1.5B through teacher-generated CoT, SFT, and Forward KL. Workflow: vLLM teacher generation, reward/grader parsing, cleaning, train/val split, benchmark evaluation, and summary tables. Result: 40,300 teacher traces; macro Pass@1 on AIME 2024, MATH500, AMC 2023, Minerva Math, and OlympiadBench improved from 46.97 to 54.45, with a 55.90 local RL teacher.
- Multi-agent RL / NLHF / HeteroCrossPlay: led a preference-learning research track where chosen/rejected pairs are produced by interaction between multiple models rather than only by a static preference dataset. Contribution: NLHF/cross-play framing, game-theory and LLM-alignment connection, verifier-reward framing, comparison with self-play and DPO/PPO/GRPO-style approaches, and experimental setup around a modified VERL recipe.
- LLM compression / pruning: participated in a structural-redundancy research track for decoder-only LLMs and made a major contribution to framing and analyzing compression experiments: layer pruning, decoder/layer importance search, attention-head merging, baseline comparison, and quality/size trade-off interpretation.
- Optimization and SGD dynamics tracks: led and contributed to finite-step SGD dynamics, minimax/DRO optimization, Frank-Wolfe variants, and applied optimization projects. These support careful hypothesis design, baseline comparison, and stability analysis for training objectives.

Visiting Research Student | MBZUAI | Jan 2026-Mar 2026

Studied LMO-based optimization methods and acceleration schemes, including formulations connected to Muon and Lookahead-type ideas. This work supports applied LLM research through critical paper reading, assumption checks, and ablation-driven experiment design.

SGD dynamics and multi-agent RL | Collaboration with Andrei Leonidov's group | 2025-present

Worked on theoretical framing and experiments for SGD noise and finite-step behavior, plus related multi-agent reinforcement learning topics. Applied contribution: experiment code, figure generation, Hessian-basis diagnostics, and result analysis for papers.

Data Analyst Intern | Yandex | earlier role

Automated a daily pipeline for B2B contractor ticket assignment and closure analytics, wrote YQL data processing, built operational dashboards in DataLens, and used Nirvana for internal data-workflow orchestration.

Selected projects

- Qwen / DeepScaleR math post-training: 40.3k teacher-generated CoT traces, SFT and Forward KL, reward/grader parsing, cleaning, evaluation on five math benchmarks; best SFT run improved macro Pass@1 from 46.97 to 54.45; local RL teacher was 55.90.
- Ridiculous-LLM-Compression: working code for layer pruning, decoder search, attention-head merging, layer-wise perplexity analysis, baseline comparison, and downstream behavior checks.
- SGDiffusion: controlled finite-step SGD trajectories, Langevin-style comparisons, Hessian-basis diagnostics, and reproducible paper-figure generation.

Education and skills

MIPT, Applied Mathematics and Informatics: B.S. in Applied Mathematics and Physics; current M.S. candidate in the Department of Intelligent Data Analysis.

Stack: Python, C++, SQL, Linux, Git, JSONL workflows, multi-GPU server workflows; PyTorch, Hugging Face workflows, vLLM, SFT, teacher/KL distillation, reward/grader parsing, benchmark evaluation, RL; YQL, DataLens, Nirvana.

Math: convex and nonconvex optimization, stochastic processes, minimax problems, traffic assignment, SGD dynamics.

Publications and preprints

- Aligning Distributionally Robust Optimization with Practical Deep Learning Needs - accepted to a NeurIPS 2025 workshop.
- Why SGD is not Brownian Motion: A New Perspective on Stochastic Dynamics - NeurIPS 2026 submission.
- Frank-Wolfe Modifications for Equilibrium Traffic Assignment - Computer Research and Modeling.
- Stochastic Origin Frank-Wolfe for Traffic Assignment - TFN-2025; accepted to Journal of Mathematical Sciences, Series B.
- Modeling Skiers Flows via Wardrop Equilibrium in Closed Capacitated Networks - TFN-2025; accepted to Journal of Mathematical Sciences, Series B.
- Conjugate Frank-Wolfe in Machine Learning - OPTIMA; accepted to CCIS.